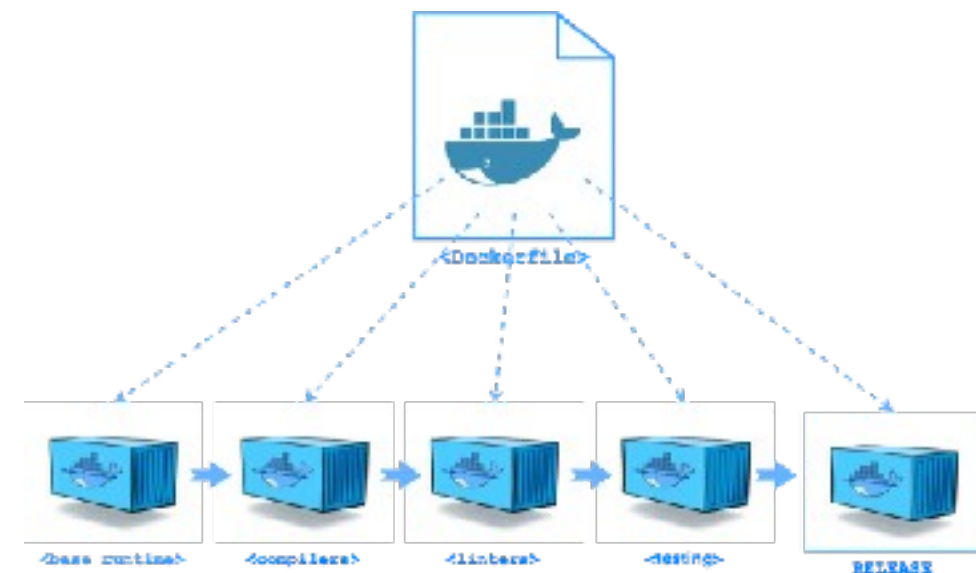
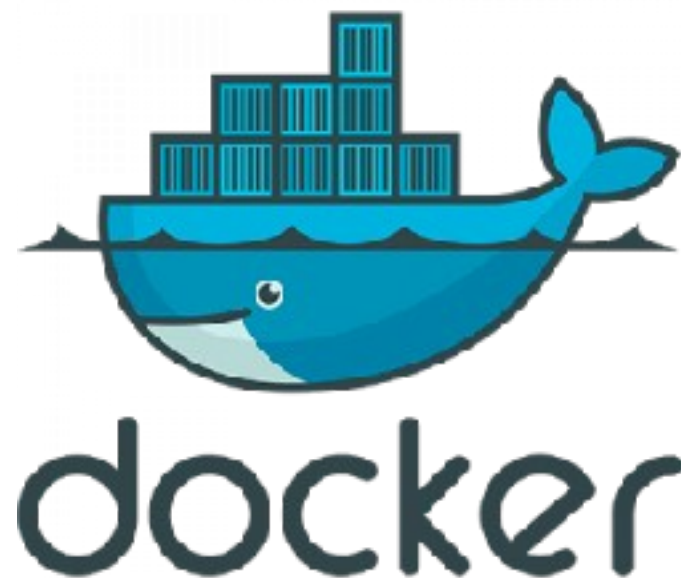


Docker multi-stage build

Using Docker multi-stage build
to create advanced build pipelines



Docker build

- **Dockerfile** - imperative DSL that defines build commands
- Each `docker build` command generates ONE image layer
- Complete **docker build** execution generates ONE Docker image

Problem with Docker Build

Image we want

runtime
configuration
application

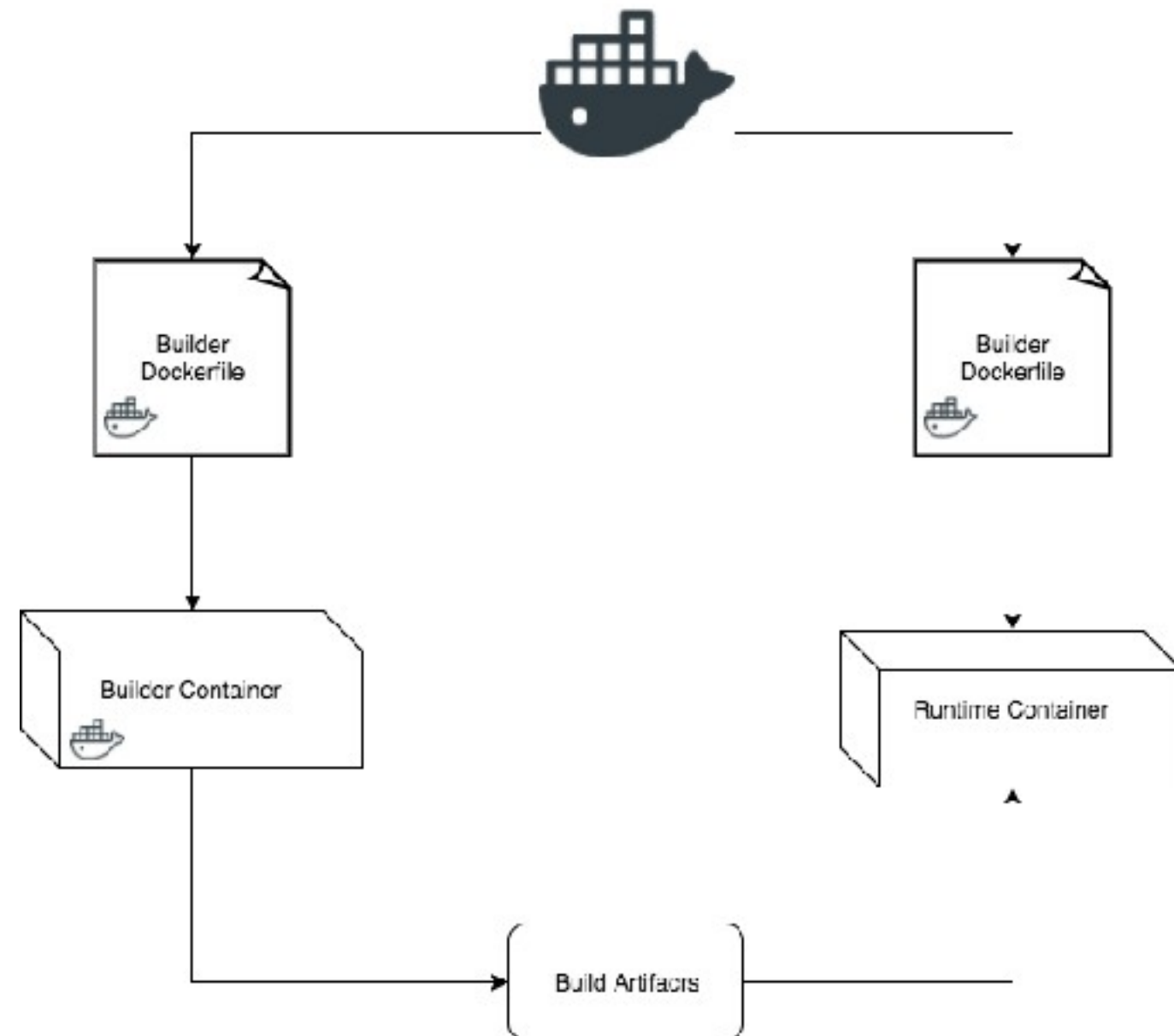
x (4..10)

Image we build

compilers, debuggers, ...
linters, tests, profilers, ...
code, build and test logs, ...
runtime
configuration
application

Docker Builder Pattern

- 2 Dockerfiles
 - 1st for build tools
 - 2nd for runtime



Drawbacks

- 2+ Dockerfiles
- Orchestration

Docker multi-stage build

✓ Benefits

- ✓ One Dockerfile
- ✓ One syntax to learn
- ✓ Same Build
 - ✓ Local and CI
- ✓ Create multiple stages

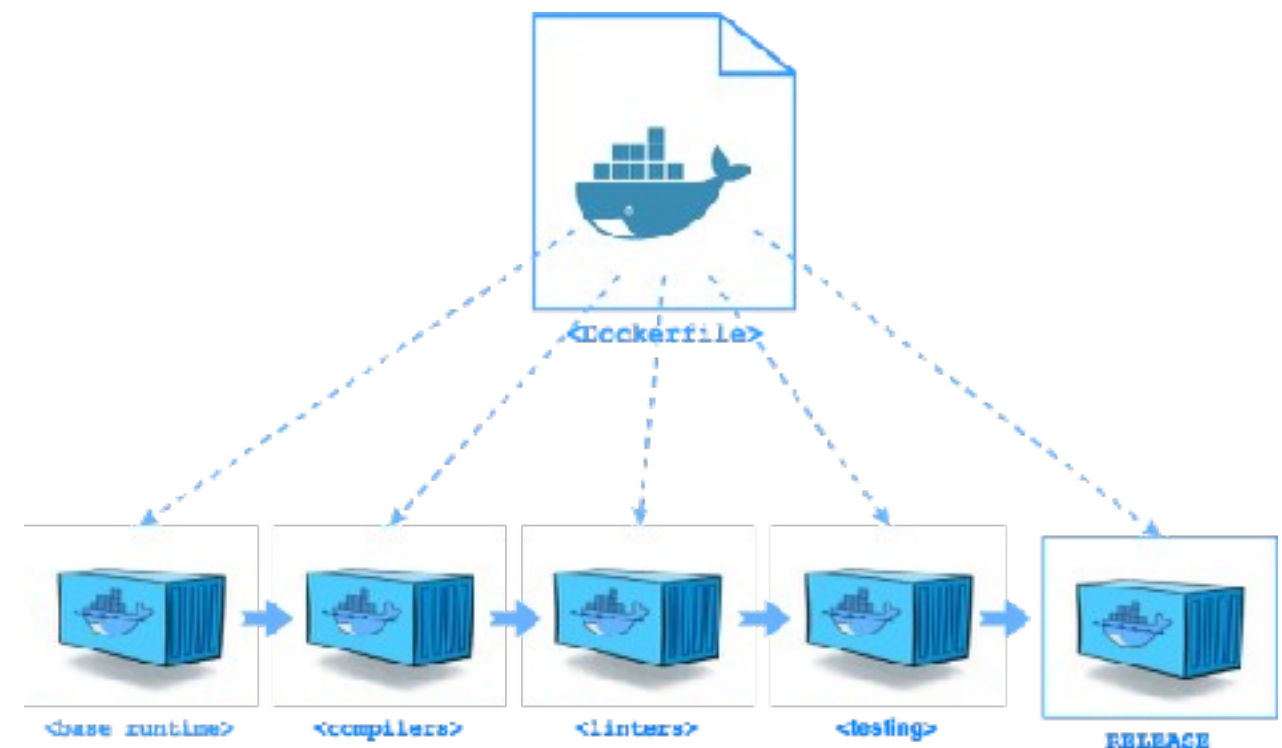
Build smaller images with Multi-stage builds

First stage: complete build environment

Second stage: minimal runtime environment ❤️



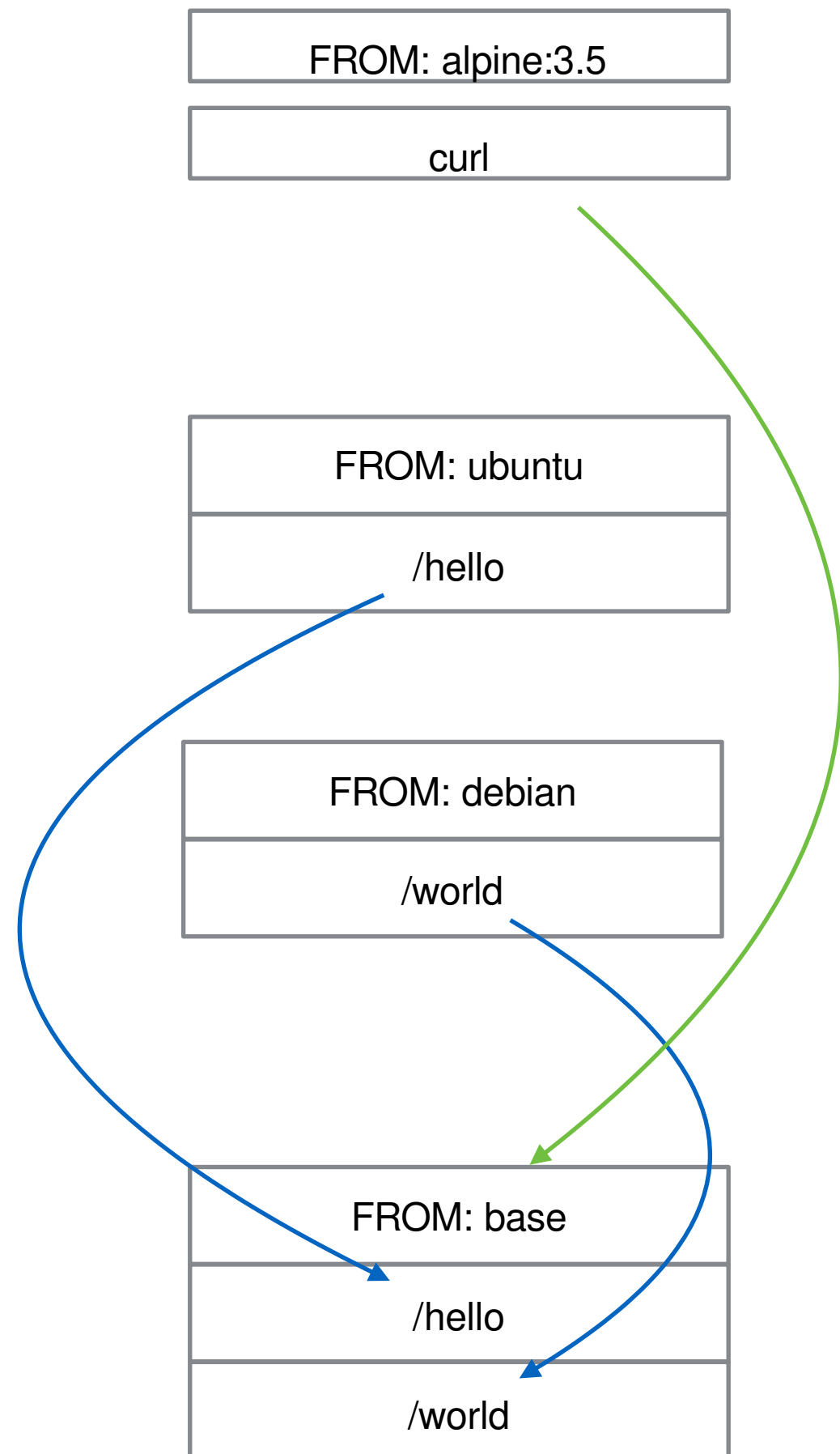
One Dockerfile, one build





Dockerfile ✕

```
1  # Base Image
2  FROM alpine:3.5 AS base
3  RUN apk add --no-cache curl
4
5  # Second Image
6  FROM debian AS second
7  RUN echo hello > /hello
8  LABEL image=second
9
10 # Third Image
11 FROM ubuntu AS third
12 RUN echo world > /world
13 LABEL image=third
14
15 # FINAL Image
16 FROM base
17 # Copy files from other images
18 COPY --from=second /hello /hello
19 COPY --from=third /world /world
20
21
```



Node.js multi-stage build

<https://github.com/codefreshdemo/demochat/tree/multistage>

Let's Chat

	Builder (Alpine)	Runtime (Alpine)
	Node.js	Node.js
Content	App Modules Yarn (package) Mocha (testing) Eslint (linter) Newman (REST testing) Gulp (automation) nsp (security scanner) Istanbul (coverage) Python gcc, g++, krb5-dev, ...	App Modules Yarn (package)
Size	568 MB	112 MB

Go multi-stage build

<https://github.com/gaia-adm/pumba/tree/multistage>



	Builder (Alpine)	Runtime (Alpine)
Content	go 1.8 glide (package) cover (coverage) go test (testing) git github-release (release) goveralls (coveralls.io) Bats (testing) Bash curl, openssl Docker CLI	pumba
Size	427 MB	15 MB

Docker multi-stage with codefresh.io

Summary

Dockerfile

Logs

Layers



gaia-adm/pumba

multistage/bb017b8

pumba

Download log

+ Initializing Process

9 seconds

✓

+ Cloning main repository...

2 seconds

✓

- multistage-build

53 seconds

✓

Removing intermediate container 31c4c10c0d91

Successfully built aa39ddbe2c93

Successfully tagged gaiaadm/pumba:dff7693a11fb613f60ae3d077a2e937c9ff1f1e6

Successfully built image: gaiaadm/pumba:dff7693a11fb613f60ae3d077a2e937c9ff1f1e6

The push refers to a repository [gcr.io/codefresh-production/gaiaadm---pumba]

Layer 'e5907091b4f3' already exists

Layer '572a8c2e054c' already exists

Layer 'c154057080f4' already exists

Layer '33e9e0cf8388' successfully pushed

Layer '3f734bc277cb' successfully pushed

Layer '14ec24a95853' successfully pushed

dff7693a11fb613f60ae3d077a2e937c9ff1f1e6: digest: sha256:e2c4ccc300fc369337c1b35e1799b233af327f440d9c139135657cc8ea283302 size: 1571

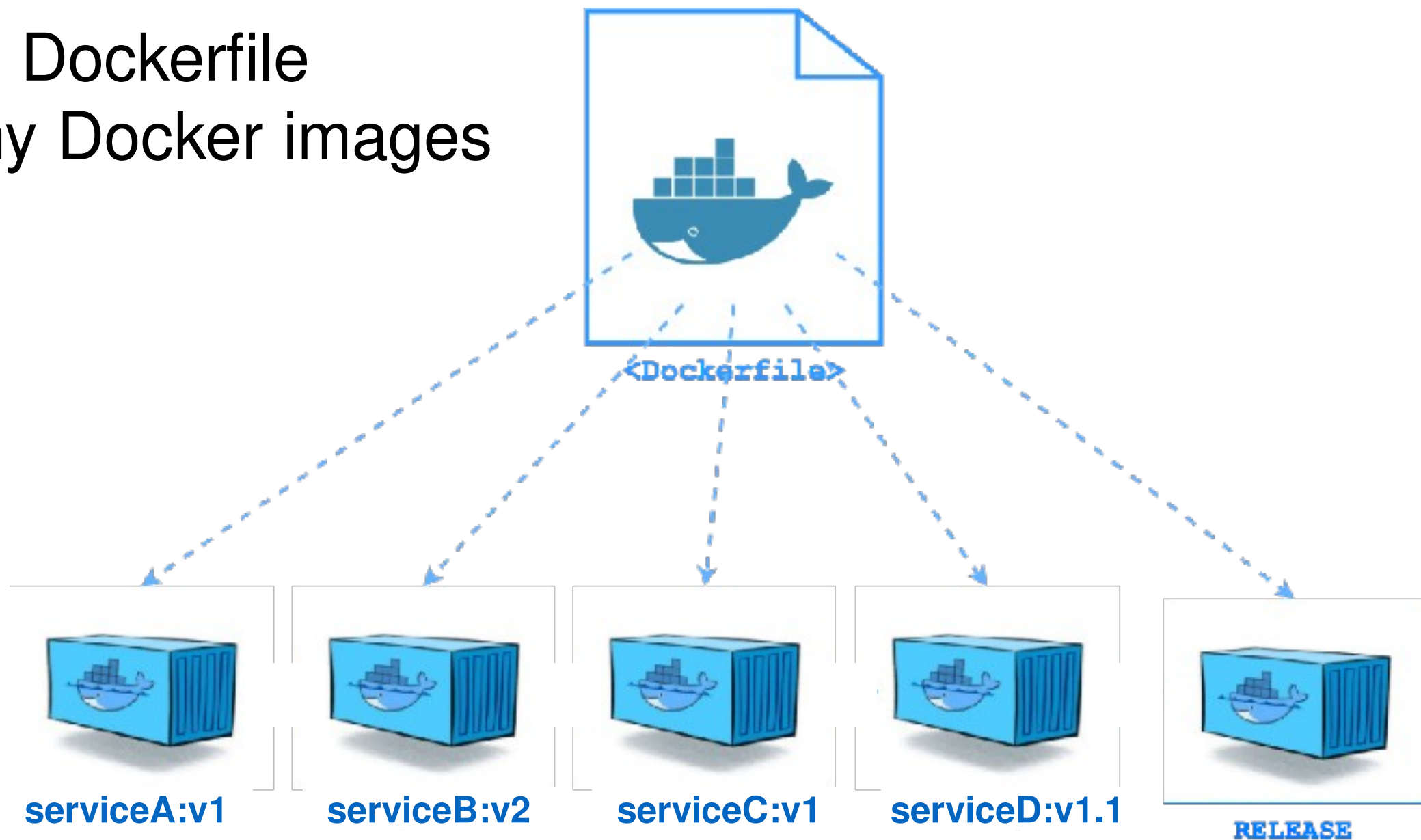
☐ Autoscroll with output

 Copy to clipboard

Completed in: ⌚ 1 min 4 sec

Multi-build Use Case

- ✓ One Dockerfile
- ✓ Many Docker images



```
$ label = "LABEL name=serviceA"
```

```
$ tag = "serviceA:v1"
```

```
$ docker tag $(docker history 224f4fee5ec1 | grep $label | awk '{print $1}') $tag
```

Docker Builder Pattern

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https://codefresh.io/blog/node_docker_multistage/